

The following problem involves the number a_3 , which is given by a computer code that represents a_3 as a computable number.

The code for a_3 is:

```
"Compute_a_3(n)
Input : natural number n
Output = 1 + (1/2)^n - 72^(-n)
end
"
```

Is it the case that the number a_3 is positive?

⊠ Thought for 24 seconds >

Yes, the number a_3 is positive.

The code defines a_3 by means of a computable sequence of rational approximations: on input n , it outputs $1 + \left(\frac{1}{2}\right)^n - 72^{-n}$. As $n \rightarrow \infty$, both $\left(\frac{1}{2}\right)^n$ and 72^{-n} tend to 0, so the sequence converges to 1. Hence $a_3 = 1$, which is clearly positive. Even without taking the limit, every output value is strictly greater than $1 - 72^{-1} > 0$.